

PHASE 1

Brainstorming and Ideation Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

Project Focus
AI-assisted learning resources, secure backend services, and personalized study support.

Phase Objective

To identify the learning problems the proposed system should address and transform those observations into a practical AI-based project concept.

1.1 Project Idea

The initial concept is an AI-powered study companion named AI StudyBuddy. The platform is intended to reduce the manual effort involved in preparing revision material by using generative AI to transform student-provided learning content into useful study resources.

1.2 Observed User Problems

- Students commonly work with material spread across lecture notes, PDFs, textbooks, online resources, and handwritten notes.
- Long study documents can be difficult to review quickly when examinations are approaching.
- Preparing summaries, revision cards, and practice questions manually takes considerable time.
- Students may need different study schedules depending on their available time and examination dates.
- Using separate tools for notes, quizzes, and planning can make the overall study process less organized.

1.3 Proposed Capabilities

- Secure student registration and login.
- Upload or provide study material as input.
- Generate short summaries through an AI service.
- Create question-and-answer flashcards for revision.
- Produce multiple-choice quizzes for self-assessment.
- Create study plans based on learning needs and examination timelines.
- Store generated resources so they can be revisited later.

1.4 Technology Ideas Considered

- Node.js and Express.js for the backend and REST API layer.
- MongoDB with Mongoose for data storage and schema management.
- JWT authentication and role-based access control for protected resources.
- bcryptjs for password protection.
- Google Gemini API for generative AI operations.
- React.js as the client application that consumes the backend APIs.

1.5 Initial Outcome

The brainstorming stage resulted in a clear project direction: build a centralized learning platform that combines secure backend services with generative AI so students can spend less time preparing study resources and more time learning.